

# CU Online — Chandigarh University

MCA Final Year Project — Milestone Report

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## MILESTONE 1 REPORT

### Study Pricing Models of AWS, Azure, and GCP

Understanding cost structures, discount mechanisms, and free-tier offerings across the three major cloud providers

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**Project Title: Multi-Cloud Cost Comparison and Optimization Analytics Dashboard**

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## **Milestone 1: Study Pricing Models of AWS, Azure, and GCP**

The milestone of the project required a deep-dive study of the pricing structures of the three major cloud providers — Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). This milestone forms the knowledge foundation upon which all subsequent data simulation, analysis, and recommendation work is built. Without a clear understanding of how each provider prices its Compute, storage, and network services, and what discount mechanisms are available, meaningful cost comparison is impossible.

### **1.1 AWS Pricing Model Study**

Amazon Web Services operates the most complex and mature pricing structure of the three providers, reflecting its position as the oldest and most extensively used cloud platform. AWS pricing is published at the level of individual SKUs (Stock Keeping Units), with prices varying by service type, instance family, region, operating system, and tenancy model.

On the compute side, EC2 (Elastic Compute Cloud) is the foundational service. On-demand pricing requires no commitment and charges per second (minimum 60 seconds). Reserved Instances require 1-year or 3-year commitments and deliver discounts of 30 to 72 percent depending on payment upfront terms. Savings Plans offer similar discounts with more flexibility — the commitment is to a dollar-per-hour spend rate rather than a specific instance configuration. Spot Instances allow access to unused EC2 capacity at discounts of 50 to 90 percent below on-demand rates, with the risk of interruption.

On the storage side, S3 (Simple Storage Service) charges per GB per month for storage, per PUT/GET request, and per GB for data retrieval and egress. The free tier for new accounts includes 5 GB of S3 storage for 12 months only.

Network egress from AWS to the internet is charged at \$0.09/GB for the first 10 TB/month from US East regions. Data transfer between AWS services in the same region is free. Inter-region transfer is charged at reduced rates.

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## 1.2 Microsoft Azure Pricing Model Study

Azure's pricing structure parallels AWS in its general architecture but differs significantly in its discount mechanisms. Azure VMs are priced on-demand at per-minute billing. Reserved VM Instances (1-year or 3-year) deliver 35 to 65 percent discounts. Azure Spot VMs provide access to unused capacity at up to 90 percent discount, similar to AWS Spot.

Azure's most distinctive pricing feature is the Hybrid Benefit: customers with active Software Assurance coverage for Windows Server or SQL Server licenses can use those licenses in Azure VMs, effectively eliminating the operating system licensing component of the VM price — a 40 to 60 percent cost reduction for qualifying Windows and SQL workloads. This benefit is unique to Azure and represents a genuine, substantial advantage for organisations with existing Microsoft licensing investments.

Azure Blob Storage charges per GB per month, per operation, and for egress. The always-free tier includes 5 GB of LRS (Locally Redundant Storage) Blob Storage with no time limit, but operation charges apply. Egress to the internet is charged at \$0.087/GB for the first 10 TB from North American regions.

## 1.3 Google Cloud Platform Pricing Model Study

GCP's pricing model is the most distinctive of the three providers. Its most important feature is sustained-use discounts: automatically applied reductions for Compute Engine resources (VMs) that run for more than 25 percent of a billing month. No commitment or reservation is required — the discount accumulates automatically as usage time increases, reaching 30 percent at 100 percent monthly usage. This makes GCP's effective compute pricing lower than its published on-demand rates for any workload that runs continuously.

Committed-use contracts (analogous to Reserved Instances) provide additional discounts of 37 to 55 percent for 1-year or 3-year commitments. Preemptible VMs are GCP's equivalent of Spot Instances, available at 60 to 91 percent below standard pricing with a maximum runtime of 24 hours per instance.

GCP Cloud Storage offers the most generous always-free tier: 15 GB of Standard Storage permanently free with no time limit and no per-operation surcharges within the free tier. This is three times the AWS free tier allocation and unrestricted by the 12-month time limit that applies to AWS. Standard egress to the internet is charged at \$0.08 to \$0.12/GB depending on destination region, with 200 GB/month of free egress for data downloaded via the Google Cloud console.

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## 1.4 Comparative Analysis Summary

| Dimension               | AWS                                    | Azure                                | GCP                                         |
|-------------------------|----------------------------------------|--------------------------------------|---------------------------------------------|
| Compute Discount Model  | Reserved (1yr/3yr commitment required) | Reserved + Hybrid Benefit licensing  | Sustained-use auto-discount (no commitment) |
| Best Compute Discount   | Up to 72% (3yr, all upfront)           | Up to 65% + Hybrid Benefit           | Up to 55% committed + 30% sustained-use     |
| Free Storage Tier       | 5 GB S3 (12 months only)               | 5 GB LRS Blob (always, ops charged)  | 15 GB Standard (always free, best tier)     |
| Egress Pricing (per GB) | \$0.09 (first 10 TB, US East)          | \$0.087 (first 10 TB, North America) | \$0.08–\$0.12 (200 GB/month free console)   |
| Pricing Complexity      | Very High (1000s of SKUs)              | High (similar to AWS)                | Medium (simpler, auto-discounts)            |
| Best For                | Widest service catalogue needs         | Microsoft-stack organisations        | Analytics, continuous compute, ML           |

Table M1.1: Cloud Provider Pricing Model Comparison Summary

## 1.5 Milestone Deliverables Completed

- Documented the complete pricing structure of AWS compute (EC2), storage (S3), and network egress services, including all discount mechanisms (Reserved Instances, Savings Plans, Spot Instances).
  - Documented Azure's compute (VM), storage (Blob), and network pricing, with specific analysis of the Hybrid Benefit licensing discount mechanism.
  - Documented GCP's compute (Compute Engine), storage (Cloud Storage), and network pricing, with specific analysis of sustained-use auto-discounts and the always-free 15 GB storage tier.
  - Produced comparative analysis table identifying the key structural differences between the three providers' pricing models.
  - Established the data simulation parameters for Milestone 2, derived from the pricing analysis findings.
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